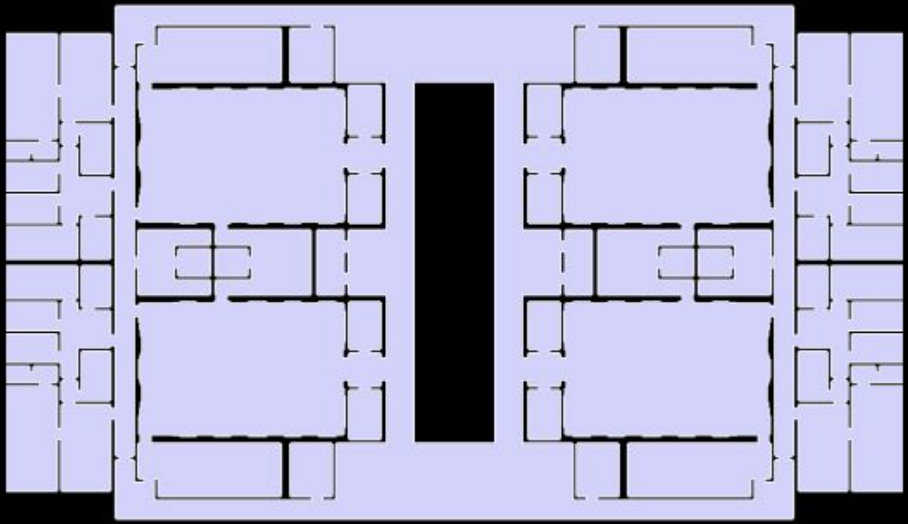
COURTHOUSE
GRAPH ANALYSIS

Graph Analysis of Architectural Structure

MaCAD · Graph Machine Learning Seminar · 2026

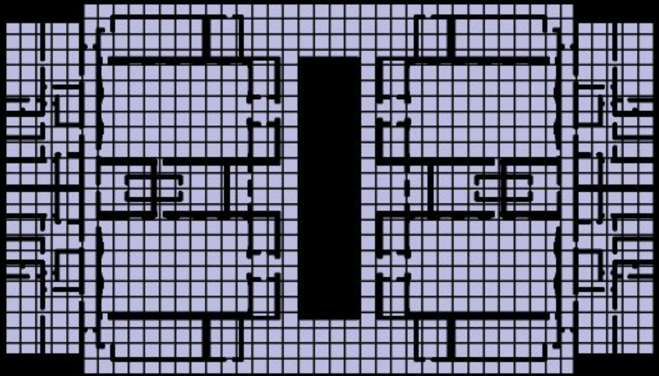
00 . Building Overview . Courthouse Floor Plan



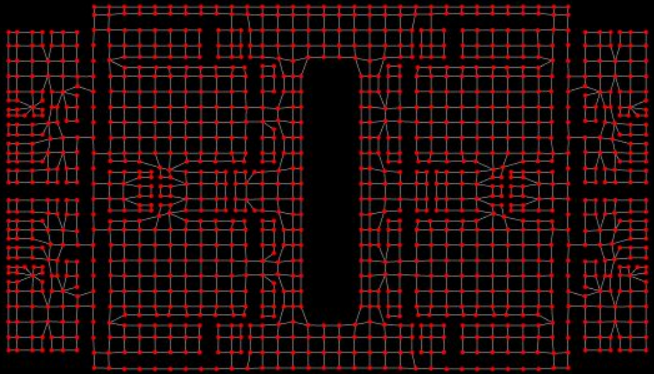
original floor plan geometry

the courthouse plan is loaded as a BREP surface and treated as a single face.

00 . Methodology . Topological Shell & Analysis Graph



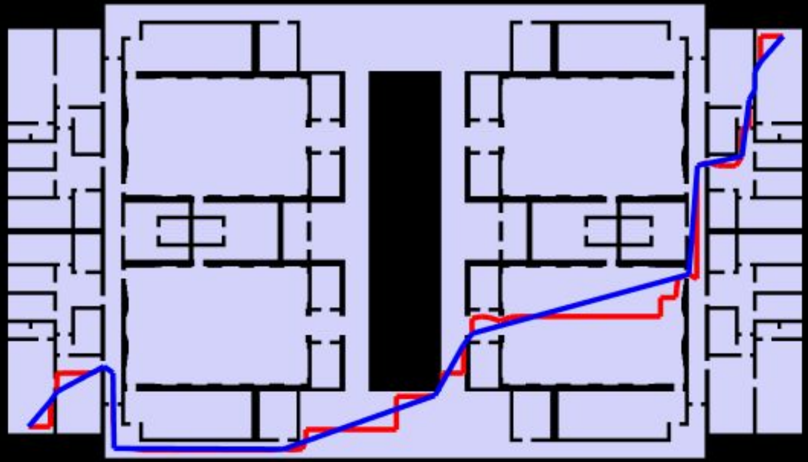
topological shell (sliced faces)



dual analysis graph

the floor plan is sliced with the grid to produce a shell of individual face cells,
each face becomes a node, shared edges become connections.

01 . Shortest Path Analysis



red = topological path · blue = straightened path

1.23 sec
Shortest Path Duration

134.38 m
Original Shortest Path Length

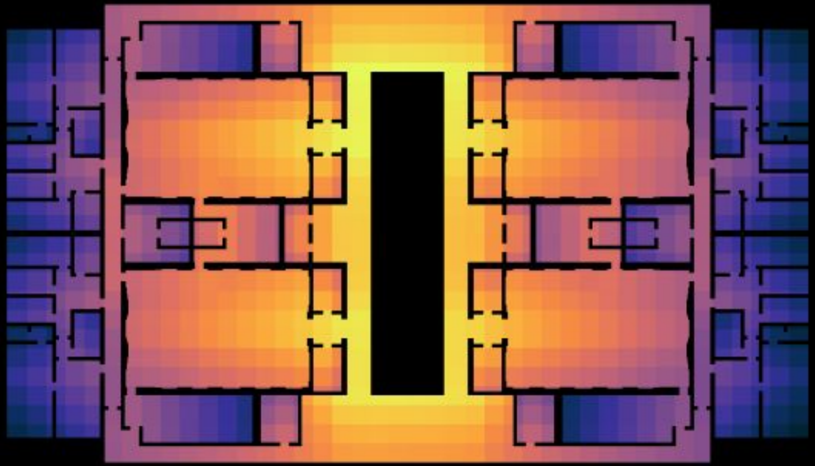
35.74 sec
Straighten Wire Duration

116.56 m
Straightened Shortest Path Length

Interpretation

The shortest walking path between extreme corners of the courthouse is 116.6 m, forced to detour around the central courtyard which acts as a circulation barrier separating north and south wings.

02 . Closeness Centrality (Global Integration)



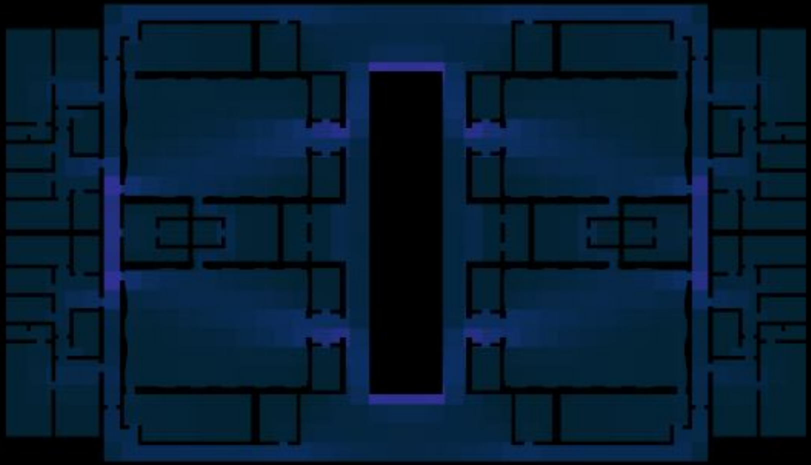
thermal scale: **cool** (low) → **hot** (high) integration

interpretation

high integration = spaces closest to all others - The ring around the courtyard scores high, marking it as the courthouse's circulation backbone.

low integration = peripheral or dead-end spaces - in a courthouse, candidates for chambers, deliberation, or other privacy-sensitive functions.

03 . Betweenness Centrality (Choice / Bottleneck)



thermal scale: cool (low) → hot (high) betweenness

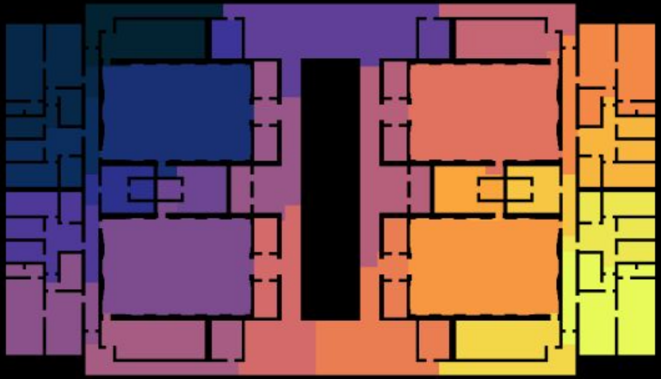
interpretation

high betweenness = critical connectors lying on the most shortest paths; removal would fragment circulation.

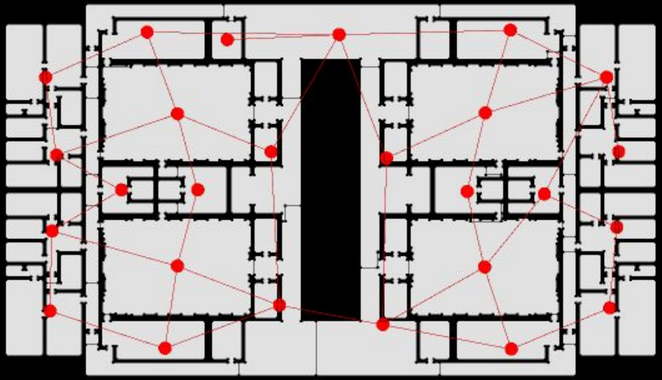
these spaces act as chokepoints or strategic transitions - in a courthouse, the doorways and corridor links that funnel cross-wing movement.

low betweenness = spaces bypassed by through-traffic, suitable for private or secure functions where uninterrupted occupancy matters.

04.1 . Community Detection (Spatial Zoning)



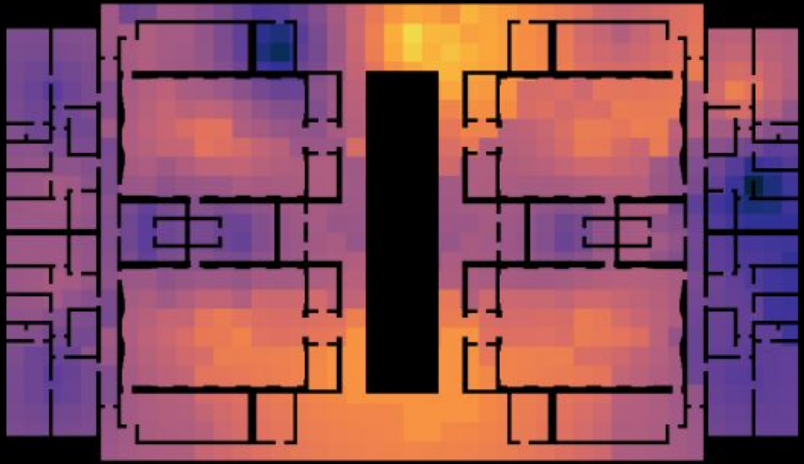
community partition (thermal color scale)



community-level graph overlay

louvain community detection groups cells by modularity maximisation, revealing natural spatial clusters without prior labelling.

04.2 . Degree Centrality (Community Connectivity)



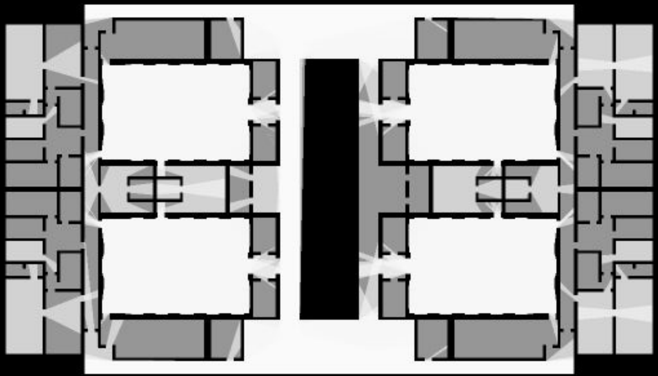
degree centrality heat-map

interpretation

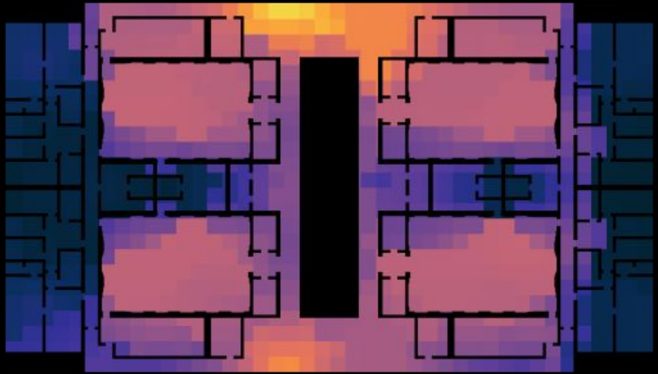
hot zones are functional hubs that bridge multiple spatial clusters.

cold zones are more self-contained or semi-private sectors of the building.

05 . Isovist & Visibility Graph



isovist grid (8m spacing)



isovist face visibility (8m spacing)

the corridors next to the courtyard have the **highest visibility** - they overlook multiple halls at once.
while the peripheral wings are the most **visually isolated**.

Key Findings & Why Graph Analysis

Key Findings

- Central corridors score highest in both closeness and betweenness, confirming their role as primary circulation spines.
- Community detection reveals 4–6 distinct spatial zones that map to functional areas (public, semi-public, restricted).
- High-degree community hubs correspond to transition spaces between public and judicial zones.
- Isovist analysis shows visibility concentrates around open atria and corridor intersections.

Why Graph Analysis?

- Encodes spatial relationships mathematically, enabling objective, reproducible analysis beyond visual inspection.
- Scales to complex buildings where manual reading is infeasible.
- Centrality metrics directly translate to architectural concepts: integration, choice, and accessibility.
- Community detection automates spatial zoning, a task that otherwise requires domain expertise.